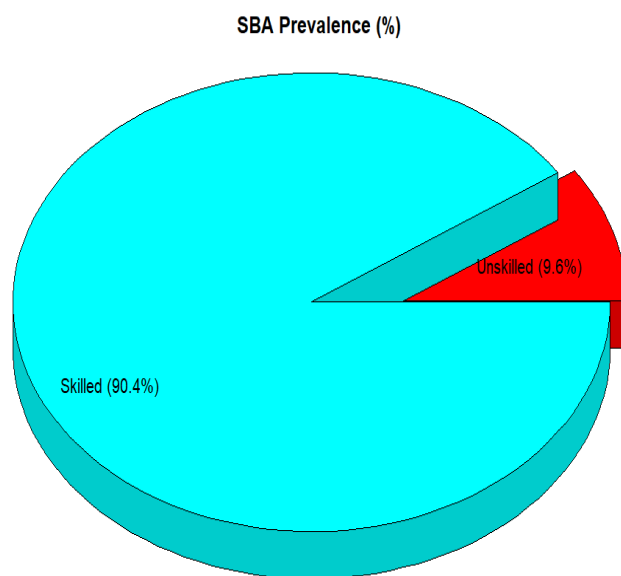
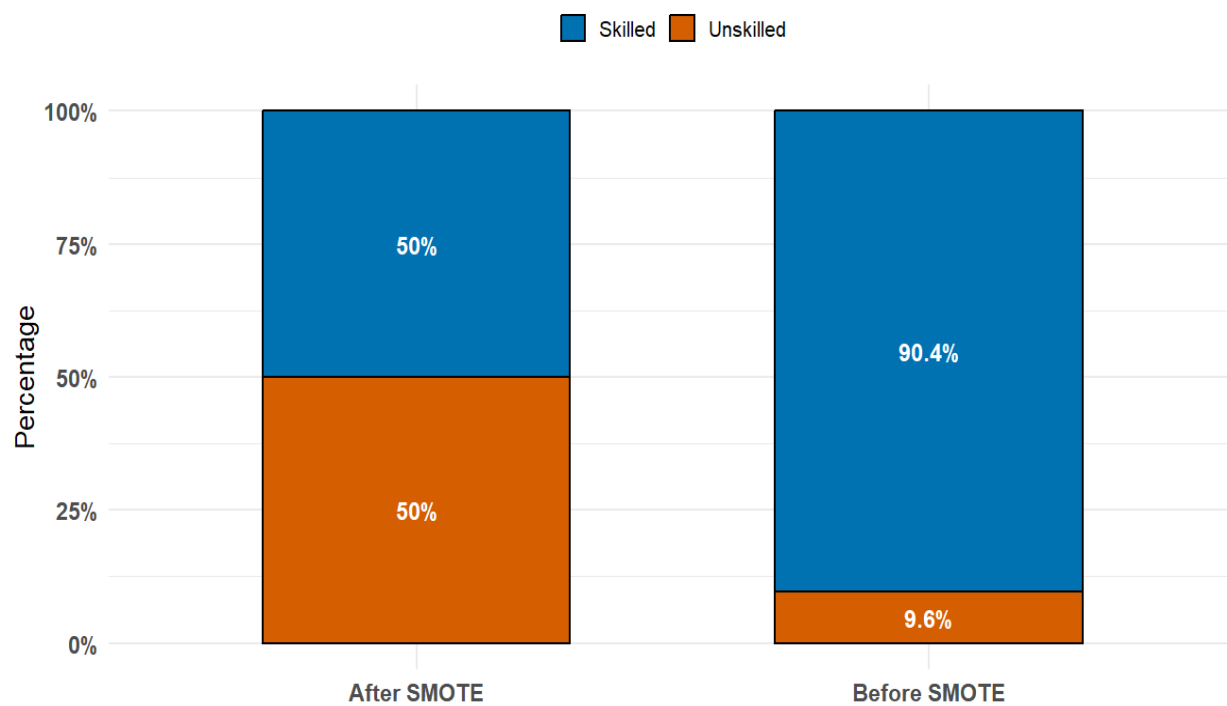
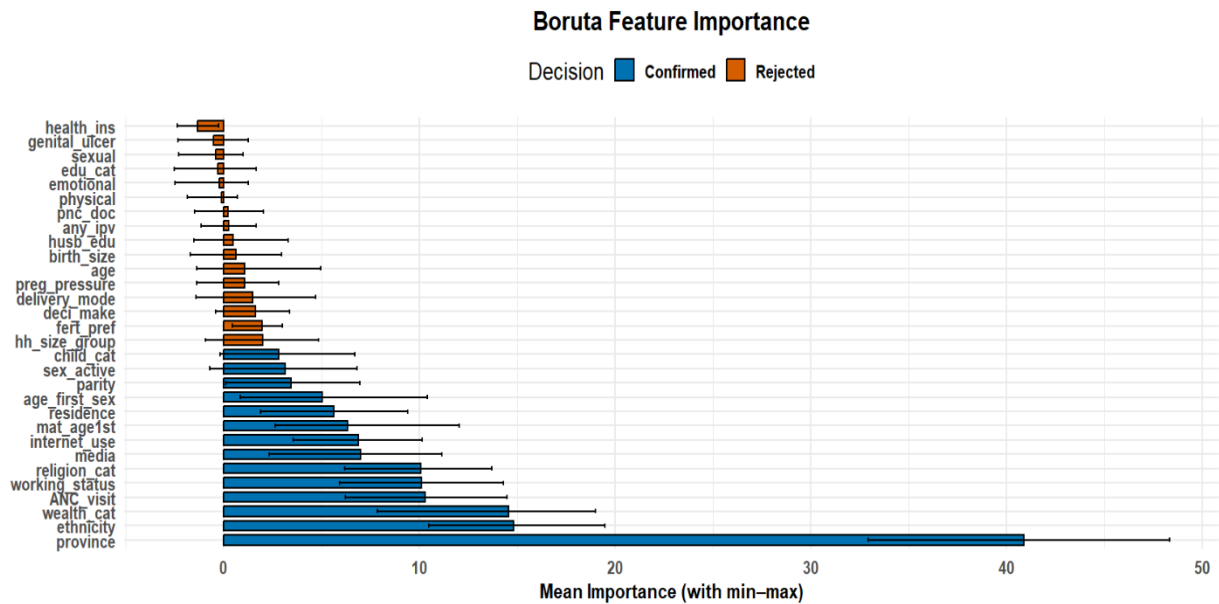




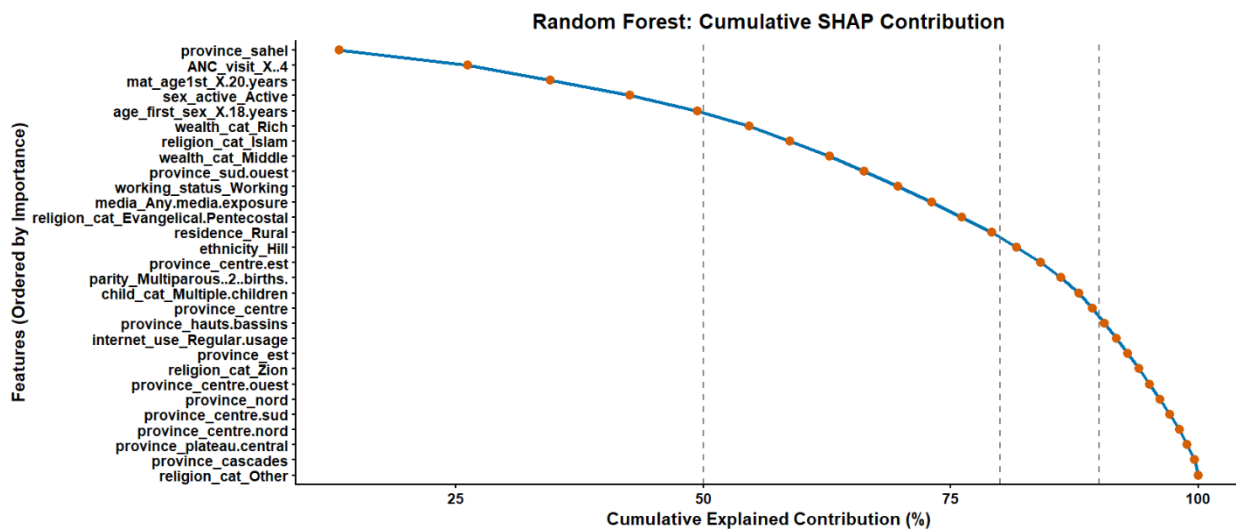
Supplementary Figure S1: Skilled Birth Attendance Status



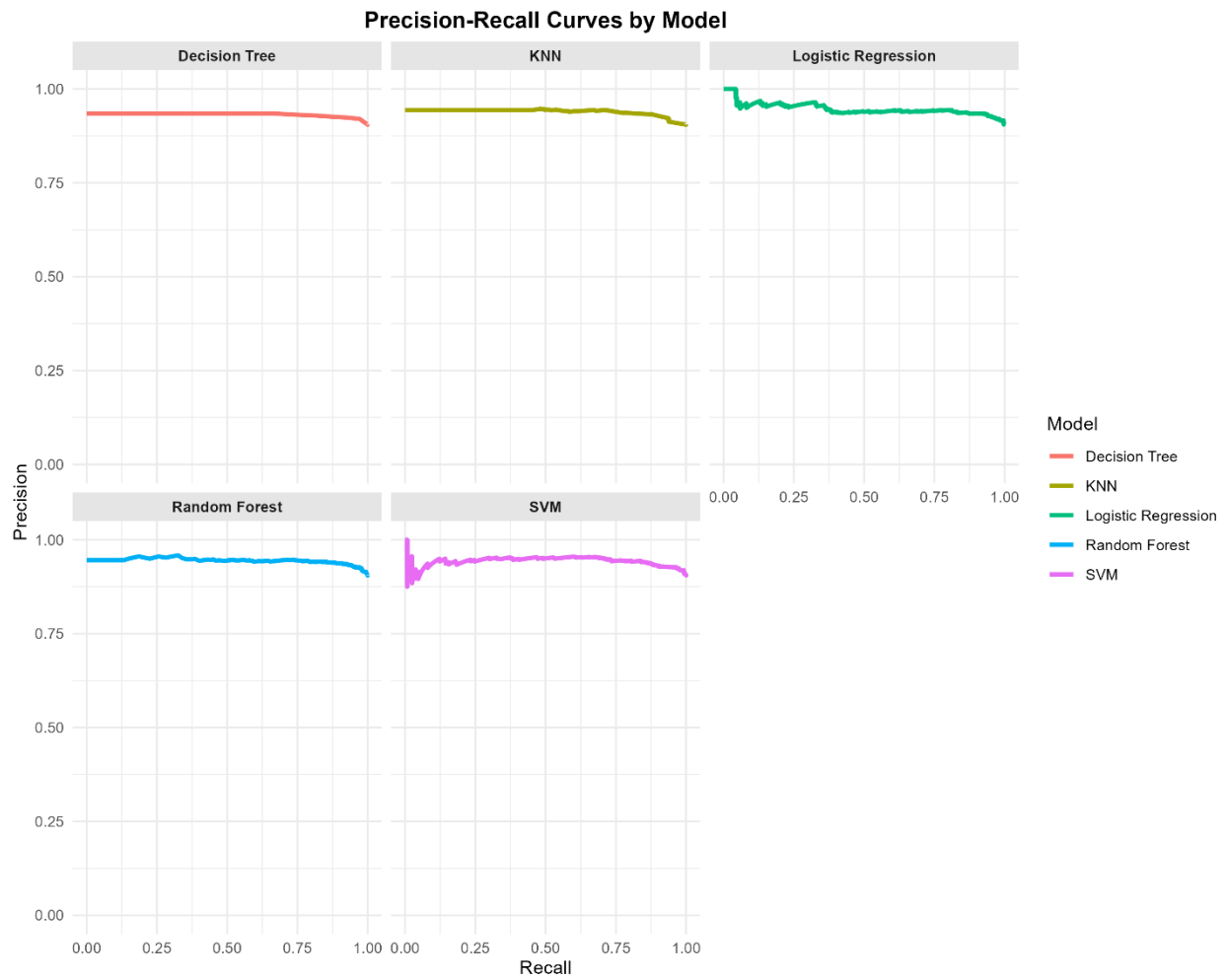
Supplementary Figure S2. Class Distribution of Skilled Birth Attendance Before and After SMOTE



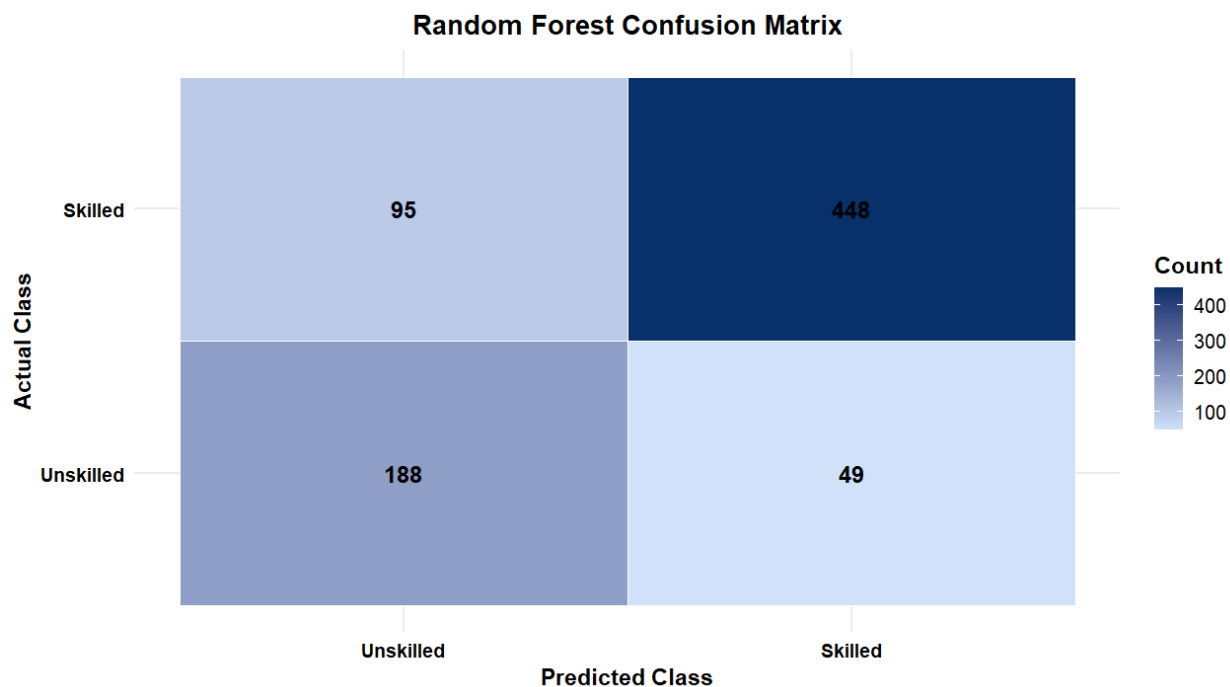
Supplementary Figure S3. Boruta Feature Importance and Selection of Predictors for Skilled Birth Attendance



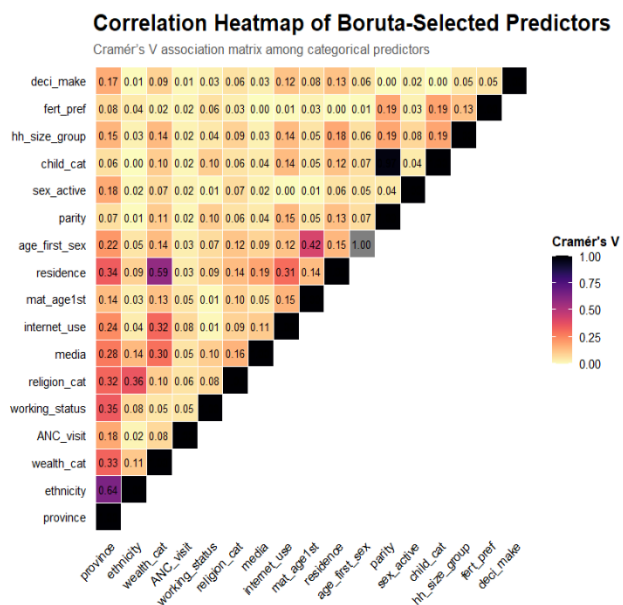
Supplementary Figure S4. Cumulative SHAP Contribution of Predictors in the Random Forest Model for Skilled Birth Attendance.



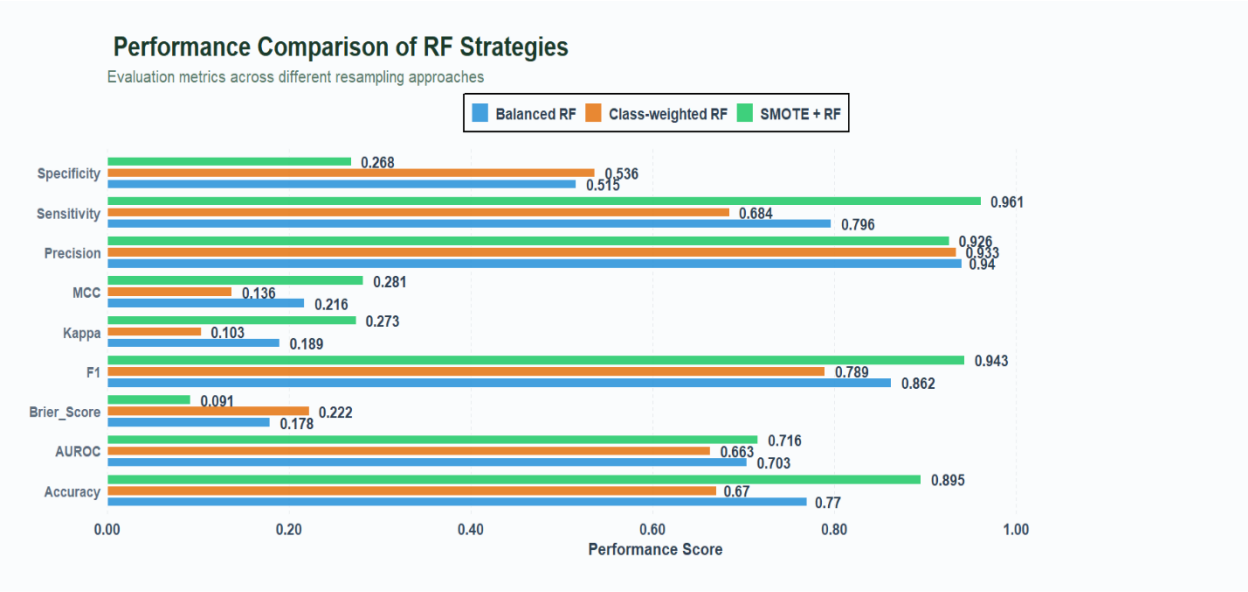
Supplementary Figure S5. Precision–Recall Curves of Machine Learning Models for Predicting Skilled Birth Attendance



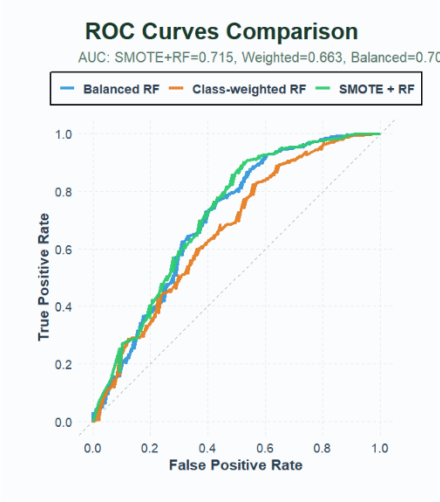
Supplementary Figure S6. Confusion Matrix of the Random Forest Model for Predicting Skilled Birth Attendance



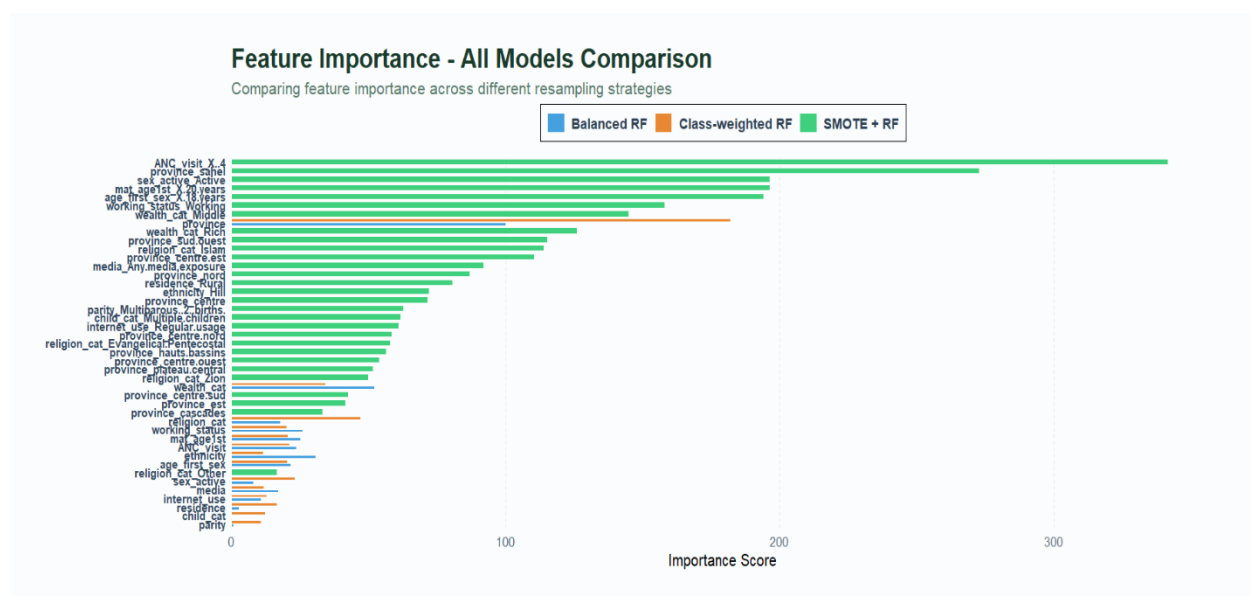
Supplementary Figure S7. Cramér's V Correlation Heatmap of Boruta-Selected Predictors for Skilled Birth Attendance



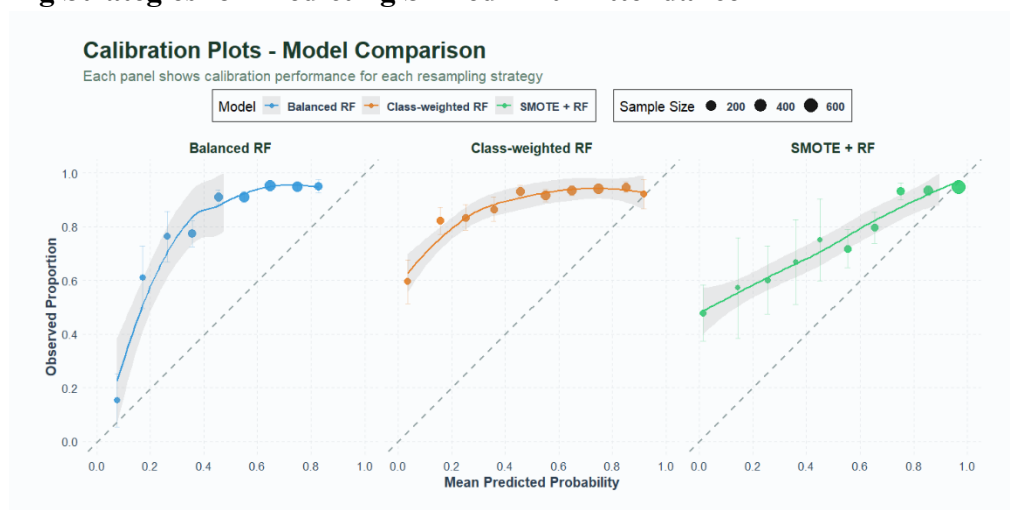
Supplementary Figure S8. Performance Comparison of Random Forest Resampling Strategies for Predicting Skilled Birth Attendance



Supplementary Figure S9. Comparison of Receiver Operating Characteristic Curves Across Random Forest Resampling Strategies



Supplementary Figure S10. Comparison of Feature Importance Across Random Forest Resampling Strategies for Predicting Skilled Birth Attendance



Supplementary Figure S11. Calibration Performance of Random Forest Models Across Different Resampling Strategies

List of Supplementary Tables

Supplementary Table S1: Descriptive characteristics of women (N = 5,111)

Variable	Category	n	%
Skilled birth attendance (SBA)	Unskilled birth attendant	489	9.57
	Skilled birth attendant	4,622	90.43
Province / Region	Boucle du Mouhoun	299	5.85
	Cascades	253	4.95
	Centre	565	11.05
	Centre Est	503	9.84
	Centre Nord	427	8.35
	Centre Ouest	477	9.33
	Centre Sud	369	7.22
	Est	299	5.85
	Hauts-Bassins	477	9.33
	Nord	399	7.81
	Plateau Central	480	9.39
	Sahel	217	4.25
	Sud-Ouest	346	6.77
Age	<20	408	7.98
	20–34	3,604	70.51
	>35	1,099	21.50
Education	No education	3,285	64.27
	Basic	777	15.20
	Secondary/Higher	1,049	20.52

Residence	Urban	1,602	31.34
	Rural	3,509	68.66
Ethnicity	Non-hill	4,506	88.16
	Hill	605	11.84
Religion	Catholic	3,370	65.94
	Islam	1,087	21.27
	Zion	372	7.28
	Evangelical/Pentecostal	216	4.23
	Other	66	1.29
Wealth	Poor	1,900	37.17
	Middle	1,072	20.97
	Rich	2,139	41.85
Child category	Single child	1,190	23.28
	Multiple children	3,921	76.72
Household size	≤5 members	1,496	29.27
	>5 members	3,615	70.73
Husband education	No education	3,266	63.90
	Basic	630	12.33
	Secondary/Higher	761	14.89
	Don't know / Missing	454	8.88
Health insurance	No	5,083	99.45
	Yes	28	0.55
Working status	Not working	1,813	35.47
	Working	3,120	61.04

	Missing	178	3.48
Parity	Primiparous (1 birth)	1,137	22.25
	Multiparous (2+ births)	3,974	77.75
ANC visits (≥ 4)	<4	1,422	27.82
	≥ 4	3,689	72.18
Delivery mode	Vaginal delivery	4,752	92.98
	Caesarean section	352	6.89
	Missing	7	0.14
Birth size	Small / Unknown	433	8.47
	Average / Large	4,678	91.53
PNC doctor	No	4,959	97.03
	Yes	152	2.97
Maternal age at first birth	<20 years	2,920	57.13
	≥ 20 years	2,191	42.87
Pregnancy pressure	Not pressured	4,607	90.14
	Pressured	250	4.89
	Missing	254	4.97
Age at first sex	<18 years	3,460	67.70
	≥ 18 years	1,651	32.30
Sexually active	Not active	1,946	38.07
	Active	3,165	61.93
Fertility preference	Do not want more	637	12.46
	Want more	4,474	87.54
Genital ulcer	No	4,737	92.68

	Yes	358	7.00
	Missing	16	0.31
Media exposure	No	572	11.19
	Yes	4,539	88.81
Internet use	Low usage	4,677	91.51
	Regular usage	434	8.49
Decision-making	No	4,150	81.20
	Yes	707	13.83
	Missing	254	4.97
Physical IPV	No	4,989	97.61
	Yes	122	2.39
Sexual IPV	No	5,076	99.32
	Yes	35	0.68
Emotional IPV	No	4,833	94.56
	Yes	278	5.44
Any IPV	No	4,770	93.33
	Yes	341	6.67

Supplementary Table S2. Model Training, Cross-Validation, and Hyperparameter Tuning Specifications

Model	Parameter	Description
All models	Cross-validation method	10-fold cross-validation was applied to assess model robustness and reduce overfitting
All models	number = 10	The dataset was divided into 10 folds; 9 folds were used for training and 1 fold for validation in each iteration
All models	classProbs = TRUE	Class probabilities were estimated to allow probabilistic prediction and AUROC calculation
All models	savePredictions = "final"	Final cross-validated predictions were saved for performance evaluation
All models	summaryFunction = defaultSummary	Standard performance metrics (Accuracy, Kappa) were computed during training
All models	verboseIter = TRUE	Training progress was displayed to monitor model convergence
Random Forest	tuneLength = 10	A grid of 10 candidate hyperparameter combinations was explored to optimize model performance
Decision Tree (rpart)	tuneLength = 10	Complexity parameters were tuned over 10 candidate values
K-Nearest Neighbors (KNN)	tuneLength = 10	The number of nearest neighbors (k) was optimized through grid search
Support Vector Machine (Radial)	tuneLength = 10	Cost and radial basis kernel parameters were tuned using a grid of 10 values
Logistic Regression	family = binomial	A binomial logistic regression model was fitted without hyperparameter tuning

Supplementary Table S3: Baseline Performance of Machine Learning Models on the Original Imbalanced Dataset (Without Class Rebalancing)

Model	Accuracy	Precision	Recall	F1_Score	MCC	Kappa	AUROC
Random Forest	0.91	0.92	0.99	0.95	0.32	0.26	0.70
Decision Tree	0.91	0.92	0.98	0.95	0.30	0.27	0.67
K-Nearest Neighbor	0.91	0.92	0.99	0.95	0.32	0.25	0.72
Logistic Regression	0.91	0.92	0.99	0.95	0.27	0.22	0.70
Support Vector Machine	0.91	0.92	0.99	0.95	0.30	0.22	0.66

Supplementary Table S4: Survey-weighted logistic regression: Odds ratios for skilled birth attendance (SBA)

Variable / Category	Odds Ratio	95% CI	p-value
Province			
<i>Boucle du Mouhoun (Ref.)</i>	<i>Reference</i>	—	—
Cascades	0.702	0.199 – 2.476	0.581
Centre	0.478	0.183 – 1.247	0.131
Centre-Est	0.226	0.083 – 0.616	0.004
Centre-Nord	0.479	0.170 – 1.351	0.164
Centre-Ouest	0.622	0.222 – 1.743	0.365
Centre-Sud	0.871	0.321 – 2.365	0.786
Est	0.268	0.089 – 0.806	0.019
Hauts-Bassins	0.478	0.170 – 1.343	0.161
Nord	1.088	0.404 – 2.932	0.867
Plateau Central	0.971	0.360 – 2.615	0.954
Sahel	0.035	0.013 – 0.092	<0.001
Sud-Ouest	0.151	0.058 – 0.392	<0.001
Age			
<i><20 (Ref.)</i>	<i>Reference</i>	—	—
20–34	1.186	0.748 – 1.880	0.468
≥35	1.498	0.811 – 2.765	0.196
Education			
<i>None (Ref.)</i>	<i>Reference</i>	—	—
Basic	1.373	0.921 – 2.047	0.119
Secondary/Higher	1.019	0.688 – 1.510	0.924
Residence			
<i>Urban (Ref.)</i>	<i>Reference</i>	—	—
Rural	0.950	0.570 – 1.581	0.842
Ethnicity			
<i>Non-Hill (Ref.)</i>	<i>Reference</i>	—	—
Hill	1.270	0.598 – 2.699	0.533
Religion			
<i>Christian (Ref.)</i>	<i>Reference</i>	—	—

Islam	0.851	0.579 – 1.250	0.410
Zion	1.023	0.592 – 1.768	0.934
Evangelical/Pentecostal	0.607	0.349 – 1.056	0.077
Other	0.564	0.197 – 1.617	0.286
Wealth			
<i>Poor (Ref.)</i>	<i>Reference</i>	—	—
Middle	1.675	1.192 – 2.353	0.003
Rich	1.357	0.869 – 2.120	0.180
Number of children			
<i>Single child (Ref.)</i>	<i>Reference</i>	—	—
Multiple children	0.795	0.320 – 1.975	0.620
Household size			
<i>≤5 members (Ref.)</i>	<i>Reference</i>	—	—
>5 members	0.688	0.509 – 0.928	0.014
Husband's education			
<i>None (Ref.)</i>	<i>Reference</i>	—	—
Basic	1.209	0.828 – 1.766	0.325
Secondary/Higher	1.272	0.871 – 1.856	0.212
Don't know/Missing	1.173	0.735 – 1.872	0.503
Health insurance			
<i>No (Ref.)</i>	<i>Reference</i>	—	—
Yes	0.638	0.087 – 4.647	0.656
Parity			
<i>Primiparous (Ref.)</i>	<i>Reference</i>	—	—
Multiparous (2+ births)	0.970	0.371 – 2.541	0.951
ANC visits			
<i><4 visits (Ref.)</i>	<i>Reference</i>	—	—
≥4 visits	1.915	1.500 – 2.444	<0.001
Birth size			
<i>Small (Ref.)</i>	<i>Reference</i>	—	—
Average or Large	1.291	0.879 – 1.896	0.193
Prenatal care provider			
<i>Non-doctor (Ref.)</i>	<i>Reference</i>	—	—
Prenatal doctor	0.682	0.289 – 1.608	0.381
Age at first birth			
<i><20 years (Ref.)</i>	<i>Reference</i>	—	—
≥20 years	1.087	0.819 – 1.444	0.562
Age at first sex			
<i><18 years (Ref.)</i>	<i>Reference</i>	—	—
≥18 years	0.900	0.622 – 1.303	0.577
Sexual activity			
<i>Not active (Ref.)</i>	<i>Reference</i>	—	—
Active	1.122	0.871 – 1.444	0.373

Fertility preference			
<i>Do not want more (Ref.)</i>	<i>Reference</i>	—	—
Want more	1.145	0.749 – 1.751	0.531
Media exposure			
<i>No exposure (Ref.)</i>	<i>Reference</i>	—	—
Any media exposure	0.973	0.708 – 1.338	0.867
Internet use			
<i>No/irregular (Ref.)</i>	<i>Reference</i>	—	—
Regular usage	0.962	0.494 – 1.875	0.910
Note. CI = confidence interval; OR = odds ratio; Ref. = reference category; SBA = skilled birth attendance.			
N = 5,111; Survey-weighted analysis; Number of strata = 26; Number of PSUs = 511; Design df = 485.			

Supplementary Table S5. Sensitivity analysis of Random Forest models using different imbalance-handling approaches

Model	Accuracy	Sensitivity	Specificity	Precision	F1-score	MCC	AUROC	Brier Score
SMOTE + RF	89.50%	96.10%	26.80%	92.60%	0.943	0.281	0.716	0.091
Class-weighted RF	67.00%	68.40%	53.60%	93.30%	0.789	0.137	0.663	0.222
Balanced RF	77.00%	79.70%	51.50%	94.00%	0.862	0.216	0.703	0.179

Note: All models were trained using the respective imbalance-handling approaches on the training dataset only and were evaluated on the original independent test dataset without any resampling. Sensitivity and specificity were calculated using "Skilled" as the positive class.